

13/09/2025

Task-8.1

Aim: To write a program for various implementation of various text file operations.

Algorithm:

- 1) write to file:
 - Define writefile(filename) function:
 - open a file named "log.txt" in write mode.
- 2) Read from a file:
 - print the content.
- 3) Execute the program.
 - Read the entire content of the file.
 - print the content.

Program:

```
def writefile(filename):  
    f = open("log.txt", "w")  
    f.write  
    f.close()  
def readfile(filename):  
    with open(filename, "r")  
        content = file.read()  
        print(content)  
writefile("write")  
readfile("text")
```

Task-8.2

Aim: You have a text log.txt containing logs of a system, write a function that counts the number of lines containing the word "ERROR".

Algorithm:

- 1) Initialize error counter
 - Initialize error_count to 0.
- 2) Open and Read file:
 - Open the file specified by filename in read mode.
- 3) Check each line for "ERROR".
- 4) Return Error count.

output:

Error objects are thrown when runtime errors occur. The error object can also be used as a base object for user-defined exceptions.

output:

Number of lines with "ERROR": 0.

5) Execute the program.

• Number of lines with 'ERROR': {error_lines}.

program:

```
def count_error_lines(filename):
```

```
    error_count = 0
```

```
    with open(filename, "r") as file:
```

```
        for line in file:
```

```
            if "ERROR" in line:
```

```
                error_count += 1
```

```
    return error_count
```

```
error_lines = count_error_lines("log.txt")
```

```
print(f"Number of lines with 'ERROR': {error_lines}")
```

```
log.txt.
```

"ERROR" objects are thrown when runtime error occurs.

Task 8.3

Aim: The report containing the details (Name, departments) of the employee in list.

Algorithm:

1) Create employee data

2) Open file with writing

3) Write employee data to file

4) Execute the program.

program:

```
def write_employee_report(filename):
```

```
    employees = [
```

```
        {"name": "Alice", "department": "HR"},
```

```
        {"name": "Bob", "department": "Engineering"},
```

```
        {"name": "Charlie", "department": "Finance"}]
```

```
    with open(filename, "w") as file:
```

```
        for employee in employees:
```

```
            line = f"Name: {employee['name']}"
```

```
            file.write(line)
```

VEL TECH	
EX NO.	
PERFORMANCE (5)	
RESULT AND ANALYSIS (5)	
VIVA VOCE (5)	
RECORD (5)	
TOTAL (20)	
DATE	

Output

Name = Alice, Department : HR

Name = Bob, Department : Engineering

Name = Charlie, Department : Finance

Task 8.5

The report contains the details (name, department) of the employees in list.

- 1) Create employee data
- 2) Read file with list of employees
- 3) Write employee data to file
- 4) Generate the program

def write_employees_report(file_name):

employees = [{"name": "Alice", "department": "HR"}, {"name": "Bob", "department": "Engineering"}, {"name": "Charlie", "department": "Finance"}]

for employee in employees:
print(f'{employee["name"]} {employee["department"]}')
print()

Example Usage
write_employee_report ("Employee_report.txt")



Result Thus, the program implement various text file operations was successfully executed and the output was verified.

VEL TECH - CSE	
EX NO.	8
PERFORMANCE (5)	5
RESULT AND ANALYSIS (3)	3
VIVA VOCE (3)	3
RECORD (4)	4
TOTAL (15)	15